BILKENT UNIVERSITY
EEE 493 – INDUSTRIAL DESIGN PROJECT

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Presentation I

19.09.2014

Outline

- Introduction
- Project Description
- Project Progress
- Usage of the End-Product

Introduction

- Project Name

  Monitoring Tilt and Temperature Data For Structural Health Monitoring
- Company
  - KAREL R&D
- Academic Mentor
  - Dr. Nail AKAR
- Company Mentor
  - Dr. Alper SARIKAN
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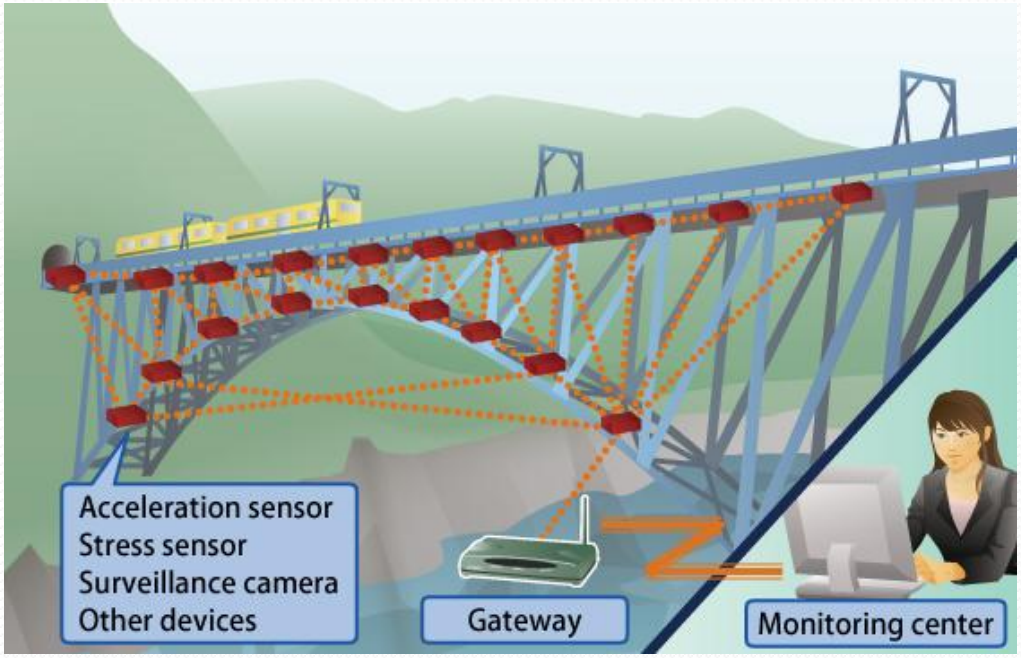
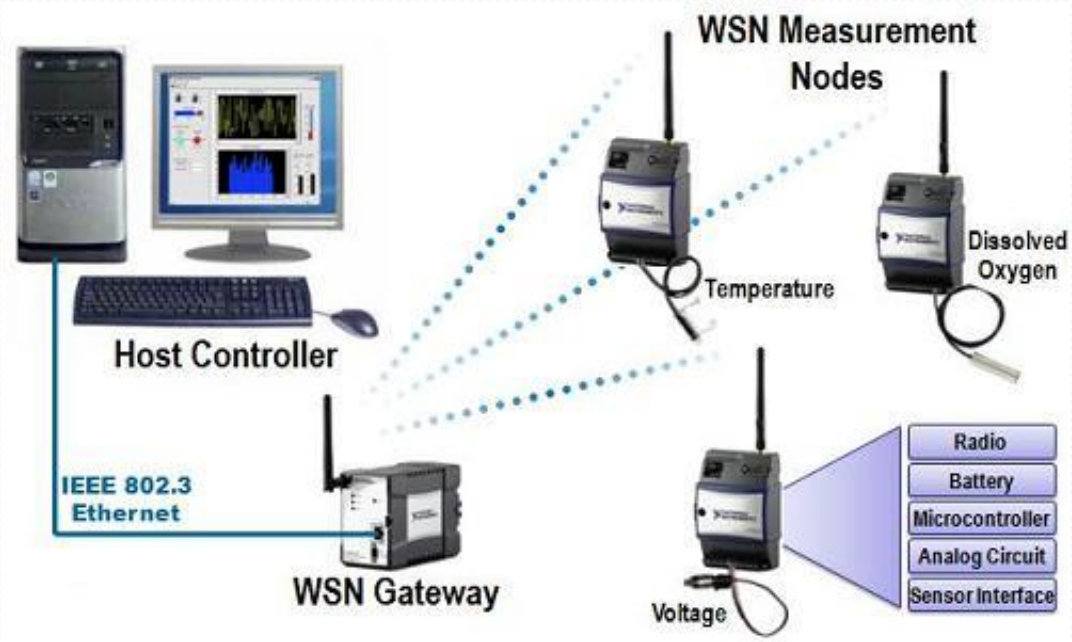
Project Description

- Structural Health Monitoring
  - Controlling Tilt and Temperature Data



Project Description

- Wireless Sensor Network
  - Autonomous sensors
  - One Sensor – One Node
  - Gateway Sensor Node (Main Machine)



Project Progress

- Previous progress
  - Preliminary research about the project
  - Meeting with company mentor

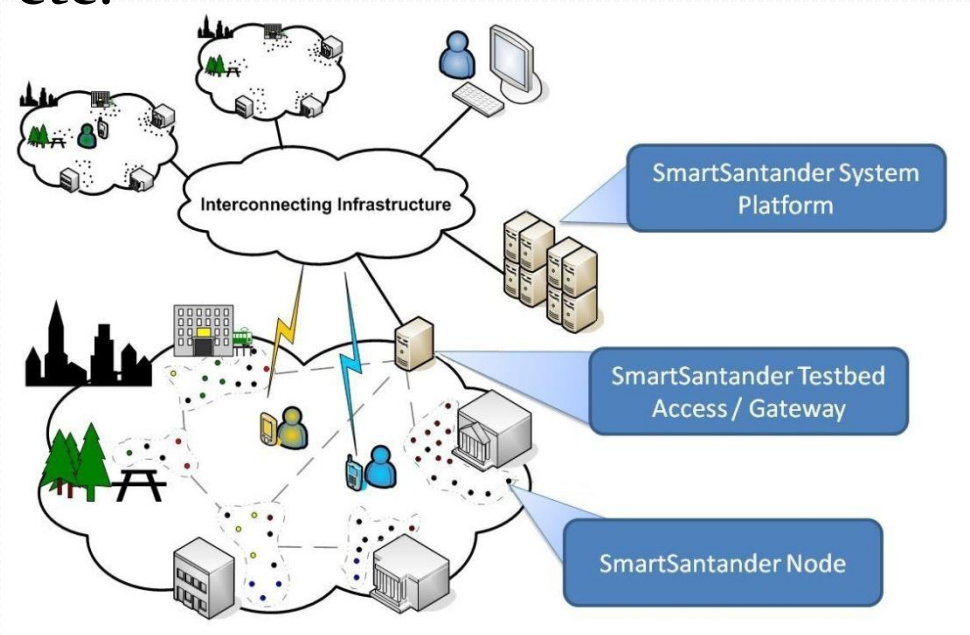
- Current progress
  - Reading articles
  - Project plan
  - Division of labor
    - System Designer, Software Designer, Electronic Designer

Project Progress

- Future Progress
  - Protocol Development
    - Header, Protection, Unique ID, Route etc.
  - Software Simulation
    - C++, QT Modeling Software
  - Analysis of the Simulations
    - Drawbacks of the protocol, corrections, route analysis
  - Hardware Design
    - Embedding developed protocol in boards
  - Measuring the target real life data

Usage of the End-Product

- Developing technology
- Smart Cities/Houses
  - Infrastructural applications
  - Bridge, Highway, Mines etc.
- Healthcare Monitoring



Thank you for listening!